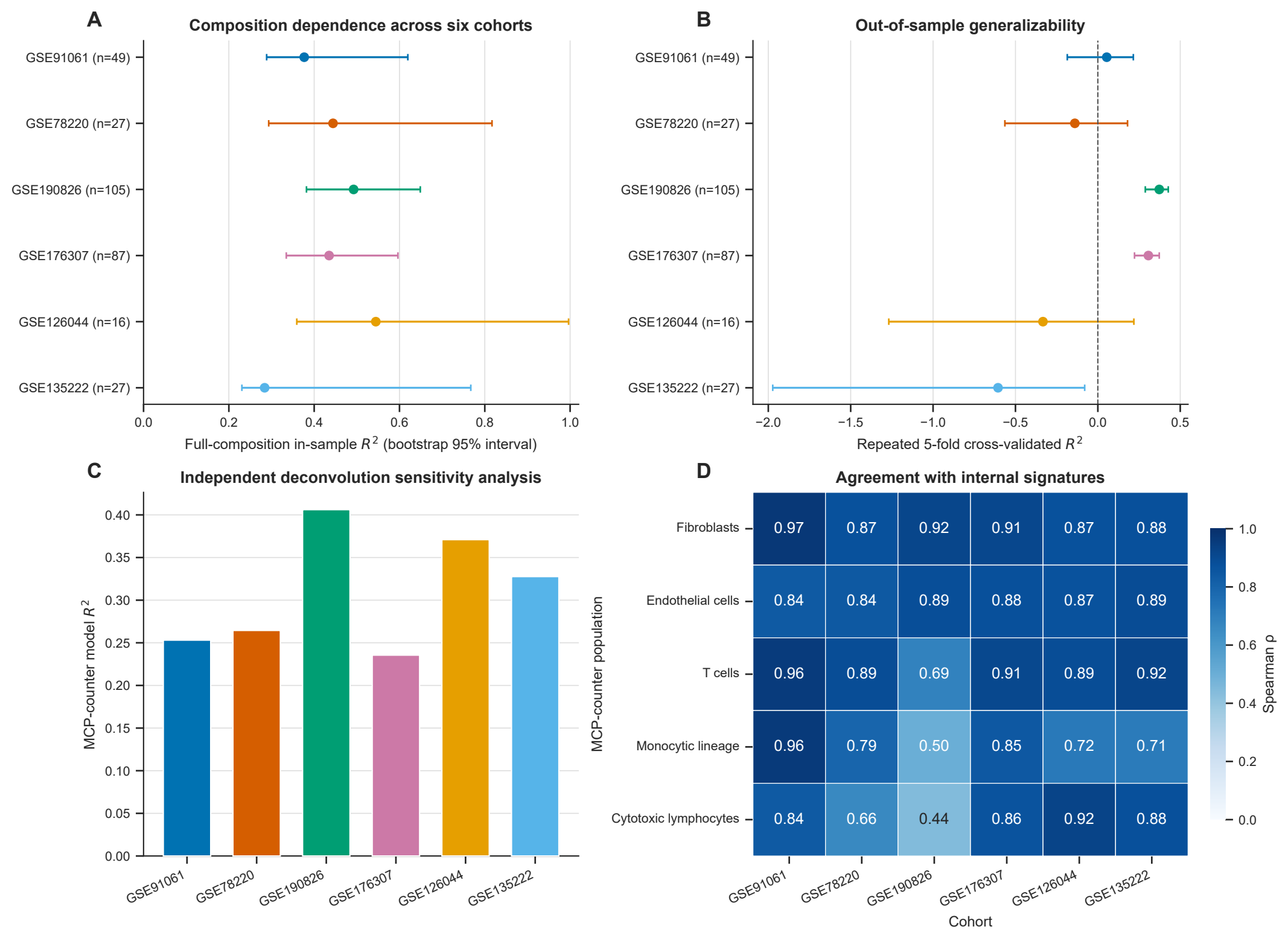

A
Composition dependence across six cohorts
GSE91061 (n=49)
GSE78220 (n=27)
GSE190826 (n=105)
GSE176307 (n=87)
GSE126044 (n=16)
GSE135222 (n=27)
0.0
0.2
0.4
0.6
0.8
1.0
Full-composition in-sample R^2 (bootstrap 95% interval)
B
Out-of-sample generalizability
GSE91061 (n=49)
GSE78220 (n=27)
GSE190826 (n=105)
GSE176307 (n=87)
GSE126044 (n=16)
GSE135222 (n=27)
−2.0
−1.5
−1.0
−0.5
0.0
0.5
Repeated 5-fold cross-validated R^2
C
Independent deconvolution sensitivity analysis
MCP-counter model R^2
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
GSE91061
GSE78220
GSE190826
GSE176307
GSE126044
GSE135222
MCP-counter population
D
Agreement with internal signatures
Fibroblasts
Endothelial cells
T cells
Monocytic lineage
Cytotoxic lymphocytes
0.97 0.87 0.92 0.91 0.87 0.88
0.84 0.84 0.89 0.88 0.87 0.89
0.96 0.89 0.69 0.91 0.89 0.92
0.96 0.79 0.50 0.85 0.72 0.71
0.84 0.66 0.44 0.86 0.92 0.88
GSE91061
GSE78220
GSE190826
GSE176307
GSE126044
GSE135222
Cohort
Spearman ρ
1.0
0.8
0.6
0.4
0.2
0.0